SET-9- Discrete-Event Simulation of Cellular System Operations with Guard Channels

1. Objective

To model a single cell of a cellular network as a discrete-event system using SimPy (Python), and to study how reserving guard channels for handoff calls affects the new-call blocking probability and the handoff dropping probability. The simulation results for 0 guard channels are validated against the theoretical Erlang B formula, and the smallest number of guard channels that keeps handoff dropping below 1% is identified.

2. Tools Used

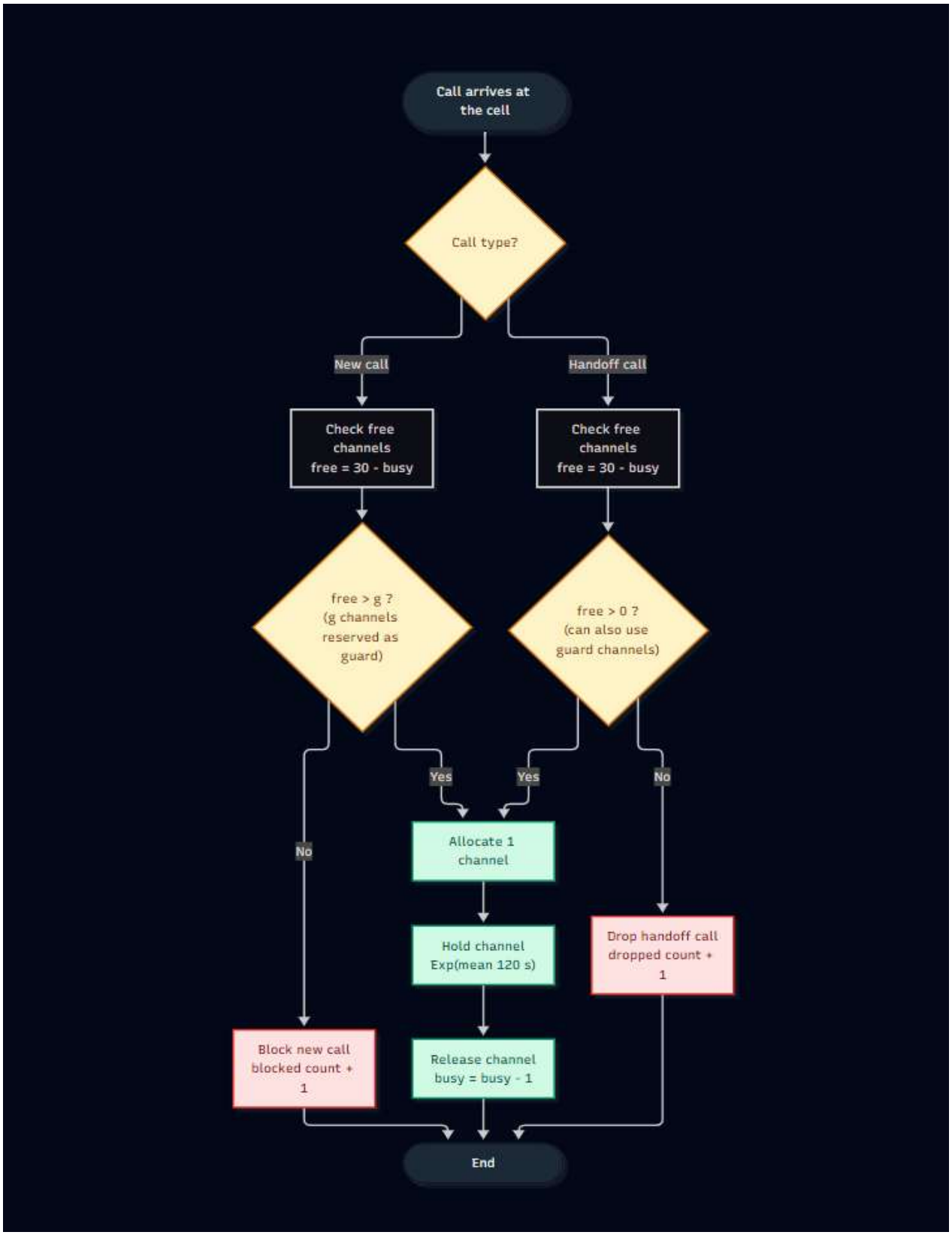
Tool	Purpose
Google Colab (Python 3)	Environment for writing and running the simulation
SimPy	Discrete-event simulation framework (event scheduling, exponential holding times)
pandas, matplotlib	Tabulating results and plotting blocking / dropping curves
Mermaid Live Editor	Drawing the call-handling flowchart (Task 1)

Channel-allocation rule

- A new call is accepted only if the number of busy channels is less than $C - g$, so the g guard channels are never given to new calls.
- A handoff call is accepted if any channel is free (busy < C), so handoff calls may use the guard channels.
- An accepted call holds one channel for an exponentially distributed time and then releases it. A call that finds no admissible channel is blocked (new call) or dropped (handoff call.

TASK-1: Draw a Mermaid flowchart of how the cell handles a new call and a handoff call

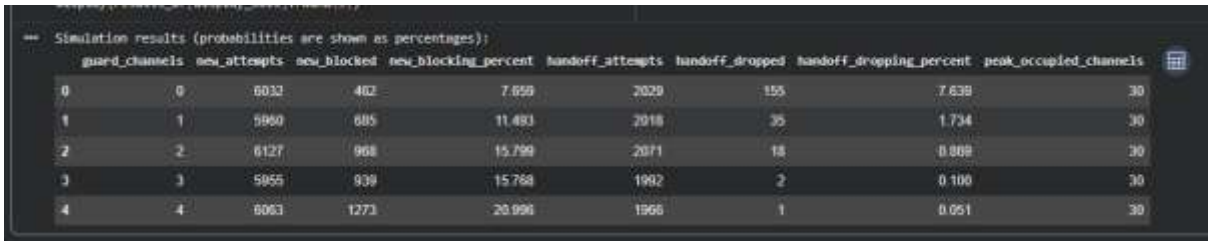
OUTPUT:



Tasks 2 and 3 — Simulate 0–4 guard channels

This is a discrete-event simulation implemented with SimPy. Each accepted call schedules a future release event. The two call types have separate arrival processes, and the channel-allocation rule gives handoff calls priority over new calls.

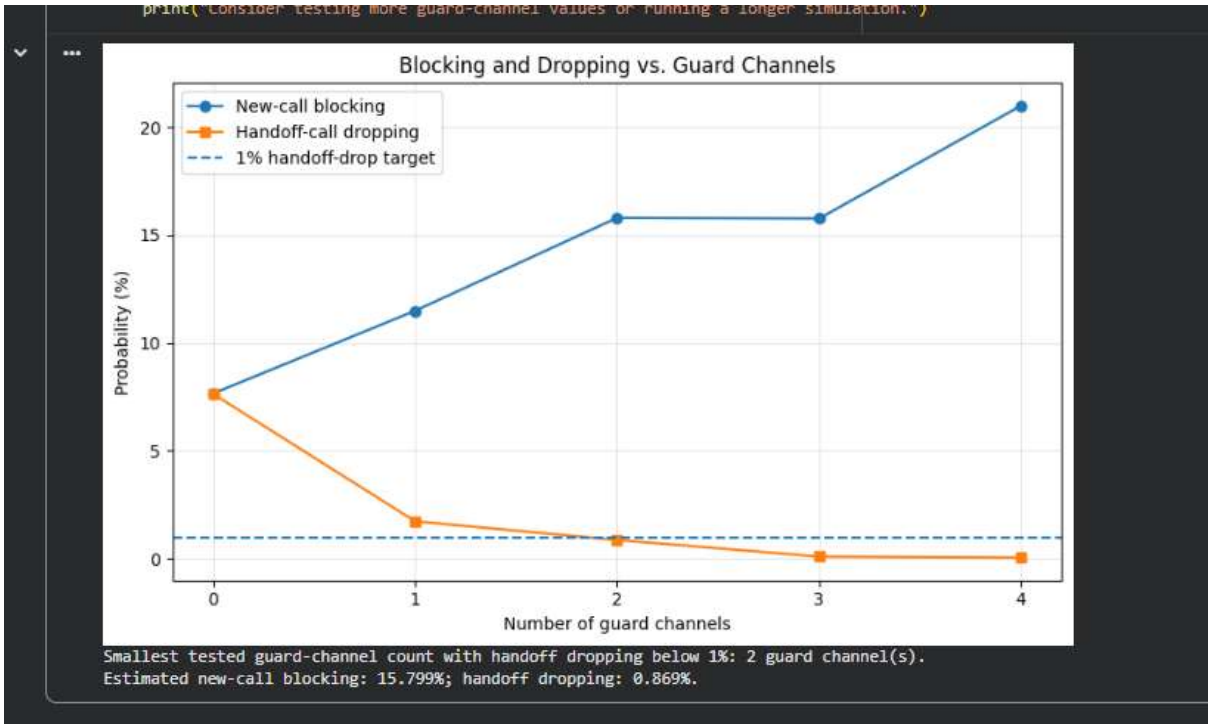
OUTPUT:



Simulation results (probabilities are shown as percentages):

	guard_channels	new_attempts	new_blocked	new_blocking_percent	handoff_attempts	handoff_dropped	handoff_dropping_percent	peak_occupied_channels
0	0	6032	462	7.659	2029	155	7.639	30
1	1	5960	685	11.493	2018	35	1.734	30
2	2	6127	968	15.799	2071	18	0.869	30
3	3	5955	939	15.768	1992	2	0.100	30
4	4	6063	1273	20.996	1966	1	0.051	30

Task 4 — Plot probabilities and check the 1% target



Task 5 — Compare the 0-guard simulation with Erlang B

OUTPUT:

index	Method	Blocking probability	Blocking (%)
0	SimPy simulation (0 guard channels)	0.07659	7.65915
1	Erlang B theoretical result	0.07667	7.66683

Total offered traffic: 26.6667 Erlangs

Absolute difference: 0.008 percentage points

Task 6 — Why is dropping an ongoing call worse than blocking a new call?

Blocking a new call is inconvenient, but the user can try again later. Dropping a handoff call disconnects an already active conversation while the user is moving between cells. This interrupts service, reduces call quality and user satisfaction, and may cause loss of important information. Therefore, cellular systems prioritize handoff calls and use guard channels to reduce handoff dropping, while balancing the resulting increase in new-call blocking.

Final observations and conclusion

Complete this section after running all cells. Use your actual generated table and plot.

1. The simulation models 30 channels, new and handoff call arrivals, and exponentially distributed call holding times.
2. Guard channels reserve capacity for handoff calls.
3. Compare the two curves to identify the smallest tested guard-channel count that brings handoff dropping below 1%.
4. More guard channels protect handoff calls but generally increase new-call blocking.
5. Compare the 0-guard simulation estimate with Erlang B and discuss the difference as a finite-simulation effect.